

# WHEN BETTER GETS WORSE: IMPROVEMENT FIDELITY FOR SELF-IMPROVING AGENTS IN ADAPTIVE WORLDS

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## ABSTRACT

Self-improving agents choose replacement operations with cheap verifiers, yet deployment can change transitions, rewards, and other agents’ behavior. We define *Improvement Fidelity* for an update  $\tau = (\pi_t, \pi_{t+1})$ : proxy improvement  $\Delta_V$  should preserve the sign and ordering of deployment improvement  $\Delta_*$ . PIVOT (Paired Interventional Validation of Optimization Transitions) estimates the differential correction  $\Delta_* - \Delta_V$  with paired high-fidelity (HF) interventions and allocates a fixed query budget by a one-step knowledge-gradient rule per unit cost.

Across controlled and sealed response worlds, we separate three phenomena that are often conflated: deployment response, update-level fidelity error, and decision-changing error. Large response can sharply increase proxy–deployment discrepancy without creating an allocation advantage. In 90 sealed Leduc, Kuhn, and Melting Pot roots, proxy and deployment top-1 updates differ in 51 roots; nevertheless, the descriptive normalized PIVOT–Uniform gain is not larger on flipped roots. The primary allocation contrast is positive in Leduc (0.036, 95% CI [0.014, 0.064]), near-null in Kuhn, and unresolved in Melting Pot, with mixed secondary conditions. A sealed-query audit finds that PIVOT’s query set intersects the proxy/deployment top-1 pair in most roots, but the queried noise relative to the proxy top-2 margin can remain large. Thus response magnitude is not decision relevance, and decision relevance is not validation value: PIVOT-KG can help when HF evidence resolves uncertainty that matters for the update decision.

## 1 INTRODUCTION

Self-improvement is usually written as a loop: generate candidates, score them, promote one, and repeat. A self-improving agent ultimately deploys a policy replacement, not an isolated policy value. In a trading system, a larger order changes liquidity; in an auction, a bid changes rivals’ responses; in a recommender, a policy changes future user behavior. A verifier can therefore rank policies correctly while ranking improvements incorrectly. High-fidelity deployment tests can expose the error, but they are often too costly to run on every candidate.

We study the transition  $\tau_t = (\pi_t, \pi_{t+1})$ . Let  $V$  be a cheap observer or replay world and let  $*$  denote the policy-induced deployment world. Define

$$\Delta_V = J_V(\pi') - J_V(\pi), \quad \Delta_* = \mathcal{J}(\pi') - \mathcal{J}(\pi). \quad (1)$$

An improvement reversal,  $\Delta_V > 0$  but  $\Delta_* < 0$ , is one salient failure mode. More generally, deployment can change the sign or ordering of candidate improvements, and a fidelity error matters to selection when it changes the selected update or displaces the no-update action. Figure 1 shows the basic sign failure.

We distinguish three response layers. The *direct* layer holds the incumbent world fixed; the *actor* layer lets the environment respond mechanically to the candidate; the *strategic* layer additionally lets other agents adapt. These correspond to progressively stronger deployment responses.

We call the agreement between proxy and deployment improvement over the updates an agent actually proposes *Improvement Fidelity*. It is operator-relative: the relevant population is the set of updates

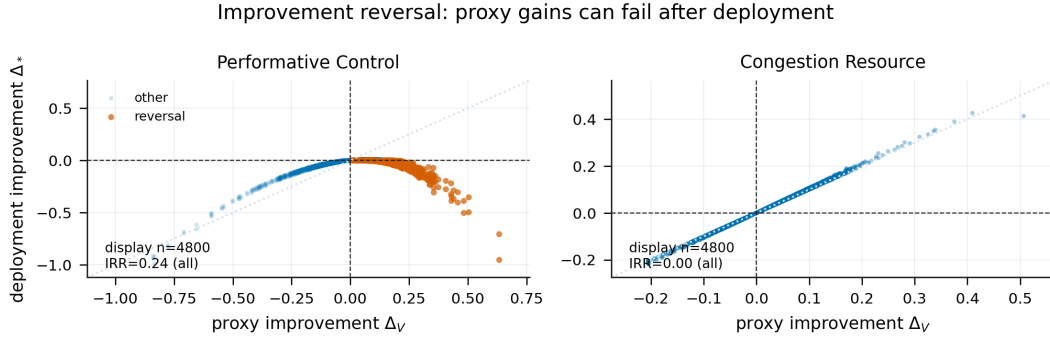


Figure 1: Paired proxy and deployment improvements in the two controlled worlds. The lower-right quadrant is an improvement reversal. Points are a deterministic display subsample; IRR is computed on all transition rows.

generated by the self-improvement process. Because only some fidelity errors change the update decision, limited high-fidelity evaluation should be targeted to uncertainty that matters for selection.

Our contributions are:

**Contribution 1.** We formalize operator-relative Improvement Fidelity for self-improving agents and show why accurate global policy evaluation can still fail on the updates an agent actually proposes. Our analysis links these failures to operator shift, deployment response, and candidate margins.

**Contribution 2.** We introduce PIVOT-KG, a paired differential validator that allocates a limited high-fidelity evaluation budget according to the expected reduction in update-selection regret per unit cost.

**Contribution 3.** We evaluate Improvement Fidelity across controlled direct, actor, and strategic response worlds and held-out decision roots, showing when response creates update-level fidelity errors, when those errors change the selected update, and when targeted high-fidelity validation reduces selection regret.

## 2 RELATED WORK

**Performative and strategic response.** Performative prediction and reinforcement learning make deployment part of the data-generating process Perdomo et al. (2020); Mandal et al. (2023); Rank et al. (2024); Mandal & Radanovic (2025); Góis et al. (2025); Piliouras & Yu (2022). These works study learning, stability, and equilibrium behavior under deployment-induced response; we ask whether a particular proposed update preserves its sign and ordering after that response is realized.

**World models and policy evaluation.** World models and policy-aware simulators evaluate values, rankings, or trajectory fidelity Quevedo et al. (2026); Dann et al. (2026); Hansen et al. (2024). Global value or ranking fidelity can coexist with a wrong local update sign, which is the gap measured here.

**Self-improvement and verification.** Autonomous policy evolution and verification make iterative self-improvement operational Wang et al. (2026); Zhang et al. (2025); Guo et al. (2026); Wu et al. (2025). We focus on a complementary failure mode: a fixed external verifier can evaluate the wrong response world, so a proposed replacement need not remain an improvement after deployment.

**Adaptive selection and paired evaluation.** Active learning, knowledge gradients, and best-arm identification allocate expensive observations Settles (2009); Lindley (1956); Frazier et al. (2009); Chick & Inoue (2001); Kalyanakrishnan et al. (2012). PIVOT changes the loss from global prediction error to update-selection regret and uses paired rollouts to cancel common-mode variation.

### 3 PROBLEM FORMULATION

**Policy-induced worlds and response levels.** Let  $\mathcal{M}[\pi]$  denote the deployment world induced by policy  $\pi$ , including any environment response to that policy. The corresponding deployment value is

$$\mathcal{J}(\pi) = J(\pi; \mathcal{M}[\pi]). \quad (2)$$

We consider three response levels:

$$\Delta_{\text{direct}} = J(\pi'; \mathcal{M}[\pi]) - J(\pi; \mathcal{M}[\pi]), \quad (3)$$

$$\Delta_{\text{actor}} = J(\pi'; \mathcal{M}[\pi']) - J(\pi; \mathcal{M}[\pi]), \quad (4)$$

$$\Delta_{\text{strategic}} = J_i(\pi', R_{-i}(\pi')) - J_i(\pi, R_{-i}(\pi)). \quad (5)$$

The direct level holds the incumbent world fixed, the actor level allows the environment to respond to the candidate, and the strategic level additionally allows other agents to adapt. Here  $R_{-i}$  denotes a general response map for the other agents; best response is one special case.

**Improvement Fidelity.** Let an improvement operator  $\mathcal{A}$  induce a transition law  $Q_{\mathcal{A}}$  over proposed updates  $\tau = (\pi, \pi')$ . For loss  $L$ , we define operator-relative Improvement Fidelity as

$$\text{IF}(V, \mathcal{A}; L) = \mathbb{E}_{(\pi, \pi') \sim Q_{\mathcal{A}}} [L(\Delta_V, \Delta_*)]. \quad (6)$$

The relevant population is therefore the updates the agent actually proposes, rather than the policy space in general. Improvement Fidelity is an update-level quantity: accurate policy values or rankings do not by themselves guarantee accurate improvement estimates.

**Core metrics.** For transition rows, we measure the magnitude of fidelity error by

$$\text{IDE} = \frac{1}{n} \sum_k |\Delta_{V,k} - \Delta_{*,k}|, \quad (7)$$

and let ISC denote sign agreement after removing ties. A salient sign failure is the improvement reversal rate

$$\text{IRR} = P(\Delta_* < 0 \mid \Delta_V > 0). \quad (8)$$

For a candidate set  $C_t$ , the decision consequence is measured by improvement-selection regret,

$$\text{ISR}_t = \max_{j \in C_t} \Delta_{*,t,j} - \Delta_{*,t,\hat{j}}, \quad (9)$$

where  $\hat{j}$  is the selected candidate. Thus IDE measures error magnitude, ISC and IRR measure sign preservation, and ISR measures the consequence for update selection.

### 4 PROPERTIES OF IMPROVEMENT FIDELITY

The following results characterize when policy-level evaluation controls update-level fidelity, how operator shift and deployment response can break that connection, and what a high-fidelity validator must resolve.

**Proposition 1** (Uniform value fidelity is sufficient). *If  $\sup_{\pi} |J_V(\pi) - \mathcal{J}(\pi)| \leq \varepsilon$ , then  $|\Delta_V - \Delta_*| \leq 2\varepsilon$ . The sign is preserved whenever  $|\Delta_*| > 2\varepsilon$ .*

*Proof.*  $\Delta_V - \Delta_* = [J_V(\pi') - \mathcal{J}(\pi')] - [J_V(\pi) - \mathcal{J}(\pi)]$ ; the triangle inequality gives the bound and the sign statement.  $\square$

**Proposition 2** (Global policy accuracy need not imply update fidelity). *For every  $\varepsilon > 0$ , there are a finite policy family, a verifier, a policy distribution, and an update operator with policy MAE and rank error below  $\varepsilon$ , but  $\text{IRR} = 1$  and  $\text{ISC} = 0$  on the operator's transition law.*

*Proof.* Take  $n$  policies with  $\mathcal{J}(i) = i/(n-1)$  and swap one adjacent pair in  $V$ . Then  $\text{MAE} = 2/[n(n-1)]$  and  $1 - \text{Spearman} = 12/[n(n^2-1)]$ , both tending to zero. Let the operator sample only the swapped directed transition and propose the lower-valued true policy. Its proxy sign is positive and its deployment sign is negative.  $\square$

**Proposition 3** (Operator Shift Bound). *Let  $P$  be a calibration law, let  $Q_A \ll P$  have density ratio  $w = q_A/p$ , and let  $\ell(\tau) = L(\Delta_V, \Delta_*)$  have finite variance under  $P$ . Then*

$$\mathbb{E}_{Q_A}[\ell] \leq \mathbb{E}_P[\ell] + \sqrt{\text{Var}_P(\ell) \chi^2(Q_A \| P)}. \quad (10)$$

*Proof.* Write  $\mathbb{E}_{Q_A}[\ell] - \mathbb{E}_P[\ell] = \mathbb{E}_P[(w - 1)(\ell - \mathbb{E}_P[\ell])]$ . Cauchy–Schwarz and  $\mathbb{E}_P[(w - 1)^2] = \chi^2(Q_A \| P)$  give the result.  $\square$

Deployment response provides a second source of update-level error. From the definitions in the previous section,

$$\Delta_{\text{actor}} - \Delta_{\text{direct}} = J(\pi'; \mathcal{M}[\pi']) - J(\pi'; \mathcal{M}[\pi]). \quad (11)$$

Thus the actor–direct gap is exactly the change in the candidate’s value caused by the world it induces. The strategic level extends this response to adaptive opponents through  $R_{-i}$ .

**Lemma 1** (Decision preservation). *For candidates  $i, j$  with true margin  $m = \Delta_{*,i} - \Delta_{*,j} > 0$ , if each estimated improvement has absolute error below  $m/2$ , then the estimated ordering of  $i$  and  $j$  is unchanged.*

*Proof.* The perturbation of the estimated margin has absolute value below  $m$ , so the estimated margin remains positive.  $\square$

**Lemma 2** (Paired variance). *For a paired rollout difference  $Y' - Y$ ,*

$$\text{Var}(Y' - Y) = \text{Var}(Y') + \text{Var}(Y) - 2 \text{Cov}(Y', Y). \quad (12)$$

*When shared randomness induces positive covariance, paired evaluation has lower variance than independent evaluation with the same marginal variances.*

Together, these results narrow the validation problem. Uniformly accurate policy values control update error, but small global MAE or rank error need not protect the transitions generated by the improvement operator. Operator shift and deployment response can therefore create local fidelity errors, while only errors large relative to candidate margins can change the selected update. Paired high-fidelity rollouts reduce the variance of estimating these differences when common randomness induces positive covariance. This motivates PIVOT-KG: allocate a limited high-fidelity budget to update uncertainty that can affect selection.

## 5 PIVOT-KG

The preceding results suggest a simple validation principle: spend expensive high-fidelity evaluations on update uncertainty that can change the selection decision. We implement this principle with PIVOT-KG. Given an incumbent and  $K$  candidate updates, the cheap verifier first provides proxy improvements  $\{\Delta_{V,j}\}_{j=1}^K$ .

Rather than model deployment improvement from scratch, PIVOT-KG models the differential correction

$$G_j = \Delta_{*,j} - \Delta_{V,j}. \quad (13)$$

At the candidate level, we place a joint Gaussian prior over the correction vector,

$$G \sim \mathcal{N}(\mu, K_{\text{prior}}), \quad \Delta_* \mid \Delta_V \sim \mathcal{N}(\Delta_V + \mu, K_{\text{prior}}). \quad (14)$$

The prior mean, covariance, and observation variances are estimated once from a disjoint calibration set of paired proxy/high-fidelity transitions. Thus a new held-out decision root starts from the calibrated posterior  $\Delta_V + \mu$ , without using its high-fidelity outcomes.

For a queried candidate  $j$ , let  $Y_j$  denote its paired high-fidelity improvement observation,

$$Y_j = \Delta_{*,j} + \epsilon_j, \quad \epsilon_j \sim \mathcal{N}(0, s_j^2), \quad (15)$$

where  $s_j^2$  is estimated from the calibration data. The incumbent and candidate are evaluated under the same initial state, exogenous randomness, and opponent initialization. Each query updates the joint posterior by Gaussian conditioning; with a non-diagonal  $K_{\text{prior}}$ , evidence can also update correlated unqueried candidates.

**Algorithm 1** PIVOT-KG under a fixed high-fidelity budget

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**Require:** Incumbent  $\pi$ ; candidates  $\{\pi'_j\}_{j=1}^K$ ; calibrated correction prior  $p(G)$ ; query costs  $\{c_j\}_{j=1}^K$ ; HF budget  $B$

**Ensure:** Selected candidate or incumbent

- 1: Evaluate each candidate with the cheap verifier to obtain  $\{\Delta_{V,j}\}_{j=1}^K$
- 2: Initialize  $D \leftarrow \emptyset$ , queried set  $Q \leftarrow \emptyset$ , and remaining budget  $b \leftarrow B$
- 3: Include the no-update action  $j = 0$  with  $\Delta_{*,0} = 0$
- 4: **while** there exists  $j \notin Q$  with  $c_j \leq b$  **do**
- 5:   Update  $p(\Delta_* \mid \Delta_V, D)$  by Gaussian conditioning
- 6:   **for** each feasible unqueried candidate  $j$  **do**
- 7:     Compute  $\text{EVSI}_j$  and  $A_j \leftarrow \text{EVSI}_j / c_j$
- 8:   **end for**
- 9:    $j_q \leftarrow \arg \max_j A_j$
- 10:   Obtain paired HF observation  $Y_{j_q}$  under matched randomness
- 11:    $D \leftarrow D \cup \{Y_{j_q}\}$ ,  $Q \leftarrow Q \cup \{j_q\}$ ,  $b \leftarrow b - c_{j_q}$
- 12: **end while**
- 13:
- 14: **return**  $a(D) = \arg \max_{j \in C^+} \mathbb{E}[\Delta_{*,j} \mid D]$

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We augment the candidate set with a no-update action,  $C^+ = \{0, 1, \dots, K\}$ , where  $\Delta_{*,0} = 0$  denotes retaining the incumbent. Let  $D$  denote the paired high-fidelity evidence collected at the current decision root. The current decision is

$$a(D) = \arg \max_{j \in C^+} \mathbb{E}[\Delta_{*,j} \mid D]. \quad (16)$$

Its posterior expected simple regret is

$$\mathcal{R}(D) = \mathbb{E} \left[ \max_{j \in C^+} \Delta_{*,j} - \Delta_{*,a(D)} \mid D \right]. \quad (17)$$

For a hypothetical high-fidelity observation  $Y_j$ , define

$$\text{EVSI}_j = \mathcal{R}(D) - \mathbb{E}_{Y_j \mid D} [\mathcal{R}(D \cup \{Y_j\})], \quad A_j = \frac{\text{EVSI}_j}{c_j}. \quad (18)$$

Thus  $\text{EVSI}_j$  measures how much querying candidate  $j$  is expected to reduce update-selection regret, while dividing by  $c_j$  favors the largest expected decision improvement per unit of high-fidelity evaluation cost. This one-step expected value of sample information gives the knowledge-gradient acquisition used by PIVOT-KG.

The primary comparison uses this fixed-budget rule; posterior-based early stopping is a secondary variant. Unlike global label acquisition, PIVOT-KG values a high-fidelity query by its expected reduction in update-selection regret.

## 6 EXPERIMENTS

### 6.1 EXPERIMENTAL PROTOCOL

The experiments answer four questions rather than treating each benchmark as a separate win report. Controlled response worlds use paired common-random-number rollouts and 30 independent seeds. The seed, trajectory, opponent-seed cluster, or held-out root is the inferential unit; transition rows are retained for audit and are not treated as independent replicates. The supplied sealed archive contains 30 held-out roots in each of Leduc, Kuhn, and Melting Pot, with 1,800 sealed decisions per cohort. We use Leduc and Kuhn poker through OpenSpiel Lanctot et al. (2019) and the Melting Pot multi-agent suite Leibo et al. (2021). Manuscript generation verifies hashes, seals, and scored-row preservation; it does not rerun an episode.

### 6.2 WHAT BREAKS IMPROVEMENT FIDELITY?

The observer/direct, actor, and strategic layers isolate progressively stronger response mechanisms. The operator-shift study is a complementary failure source: its high-minus-low shift contrast in

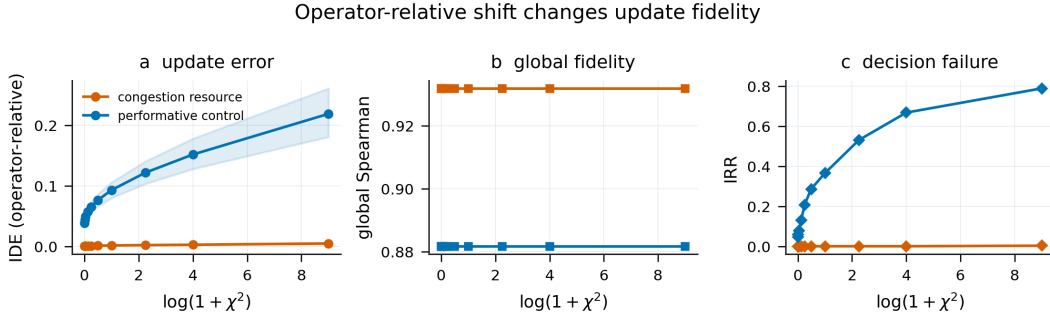


Figure 2: Operator-relative shift diagnostic. Local improvement error rises with registered shift while global policy rank remains comparatively stable. The same environment-by-shift cells show error, rank, and reversal rate.

operator-relative IDE is 0.2627 with bootstrap interval [0.1684, 0.3763] across 30 seeds and 36000 paired rows (Figure 2). Local update error rises while global policy rank remains comparatively stable. Together these results distinguish distributional operator shift from policy-induced deployment response; either can separate proxy and deployment improvement without implying that every operator or every update fails.

### 6.3 WHEN DOES FIDELITY ERROR CHANGE THE DECISION?

Lemma 1 gives the bridge: if differential error is below half the true candidate margin, the ordering is preserved. For each sealed root we therefore compute the deployment top-1 margin, the proxy-to-deployment correction, and the largest pairwise correction-to-margin ratio. A ratio at least one marks a proxy/deployment top-1 flip; it is a decision-relevance diagnostic, not an allocation outcome.

Across the 90 sealed roots, 51 have a top-1 flip (18 Leduc, 14 Kuhn, and 19 Melting Pot). To avoid comparing reward scales across worlds, Figure 3 reports each root’s squared response gap after division by the square of its largest candidate range across proxy and long/short deployment deltas. The resulting within-root normalized gaps are 0.57 [0.45, 0.70] for Leduc, 0.50 [0.38, 0.62] for Kuhn, and 0.13 [0.10, 0.17] for Melting Pot; native-unit gaps remain in the audit. Yet flip status alone does not predict validation value: normalized allocation gain is 0.075 [0.004, 0.157] on flipped roots and 0.140 [0.049, 0.250] on stable roots. Figure 3 therefore supports the sharper claim: response can create decision-relevant error, but decision relevance is not sufficient for an allocation advantage.

We then ask whether the sealed HF queries could resolve the fragile decision. For each root, the ex-post decision-critical pair is the union of the proxy and deployment top-1 candidates, while the proxy top-2 margin measures how much uncertainty separates the leading proxy candidates. At the primary long budget, the PIVOT query set intersects this pair in 0.97 [0.92, 1.00] of roots (1.00 Leduc, 1.00 Kuhn, and 0.90 Melting Pot). The descriptive normalized gain is 0.113 [0.055, 0.180] among 87 hit roots and -0.195 [-0.584, 0.000] among the three misses. The mean queried HF noise divided by the proxy top-2 margin is 13.99 [7.97, 22.90], and the final sealed estimate separation is 0.06 [0.05, 0.08] after root scaling. These are observable resolvability diagnostics, not posterior selection probabilities or an EVSI estimate: sequential posterior states are not part of the sealed decision rows.

### 6.4 CAN TARGETED HF VALIDATION REDUCE SELECTION REGRET?

Table 1 reports the sealed fixed-budget PIVOT-KG versus Uniform comparison. Leduc is the only cohort with a primary interval strictly above zero: 0.036 [0.014, 0.064]. Kuhn is near-null: 0.002 [0.000, 0.006]. Melting Pot is unresolved in the primary long/two-query condition (0.906 [-0.784, 3.401]); its one-query secondary condition is positive (3.423 [0.390, 6.944]) and its short/five-query condition is negative (-0.575 [-1.042, -0.180]). Its much larger response gap therefore does not imply a positive allocation contrast.

The closed-loop study is consistent with this scoped interpretation. Across 40 rounds, PIVOT-KG versus Proxy Only reduces paired CISR by 1.4883 [1.0618, 1.9105], while contrasts against Global-

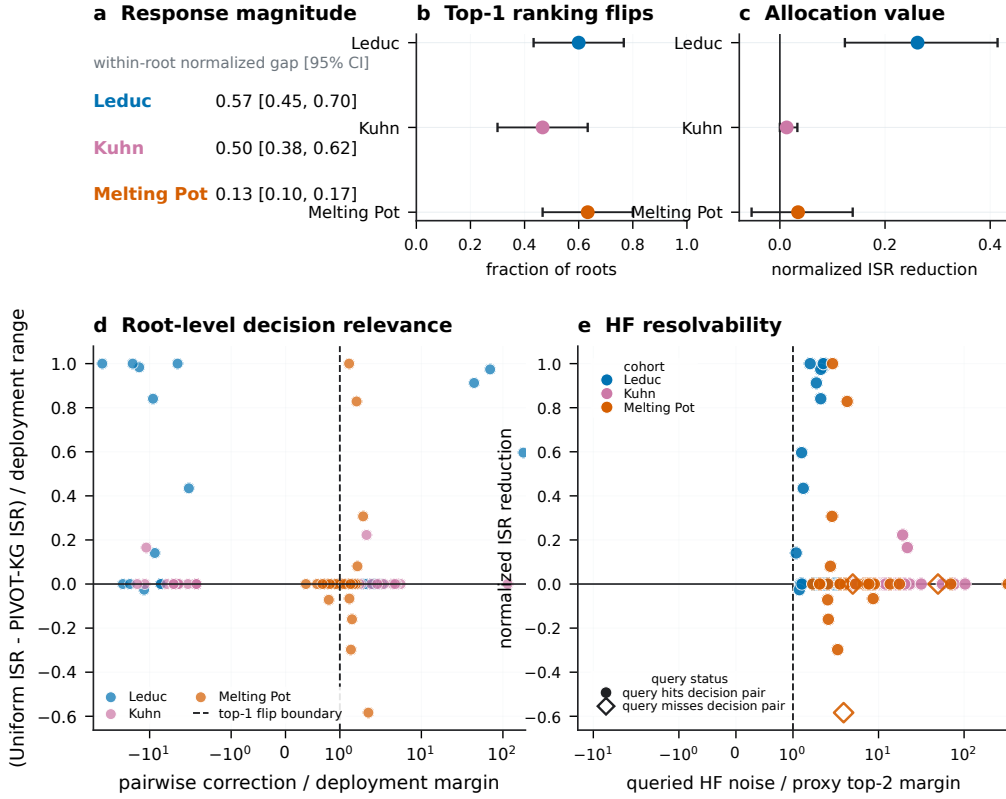


Figure 3: Response magnitude is not allocation value. a reports within-root normalized response gaps; b reports proxy/deployment top-1 flips; c reports normalized ISR reduction; d relates correction-to-margin ratio to normalized gain; e relates queried HF noise to the proxy top-2 margin, with filled circles for queries intersecting the ex-post decision-critical pair and open diamonds for queries that miss it.

Table 1: Sealed response-world allocation contrasts. Entries are paired PIVOT-KG minus Uniform gains with root-bootstrap intervals. Primary null and negative conditions are retained as evidence.

| World / condition              | $n$ | PIVOT-Uniform | 95% CI           | PIVOT ISR | Status             |
|--------------------------------|-----|---------------|------------------|-----------|--------------------|
| Leduc, long / 2 queries        | 30  | 0.036         | [0.014, 0.064]   | 0.009     | primary positive   |
| Kuhn, long / 2 queries         | 30  | 0.002         | [0.000, 0.006]   | 0.056     | near-null          |
| Melting Pot, long / 2 queries  | 30  | 0.906         | [-0.784, 3.401]  | 0.836     | primary null       |
| Melting Pot, long / 1 query    | 30  | 3.423         | [0.390, 6.944]   | 1.859     | secondary positive |
| Melting Pot, short / 5 queries | 30  | -0.575        | [-1.042, -0.180] | 1.610     | negative           |

VOI and Paired LUCB are not uniformly favorable. The evidence-efficiency frontier therefore measures a decision rule under matched costs, not universal method ordering.

## 6.5 BOUNDARY RESULTS AND NON-DOMINANCE

The powered OOD transition-evaluator comparison is a negative result rather than a superiority result: transition-minus-global ISC is -0.2590 [-0.3588, -0.1707], with the interval entirely below zero. The strategic fixture adds an independent adaptive layer and yields -0.0240 [-0.0249, -0.0231], with family-balanced SIRR 0.9495. These results show why an update-level estimand does not require a particular transition learner to dominate.

## 6.6 REACTIVE HIGHWAYENV VALIDATION

We evaluate physically reactive HighwayEnv Leurent (2018) with a fixed panel of five meta-actions (slower, lane-left, idle, lane-right, and faster). Each cohort fits PIVOT-KG on 20 calibration seeds and evaluates 60 test seeds. A budget counts candidate paired HF queries available at one decision root;

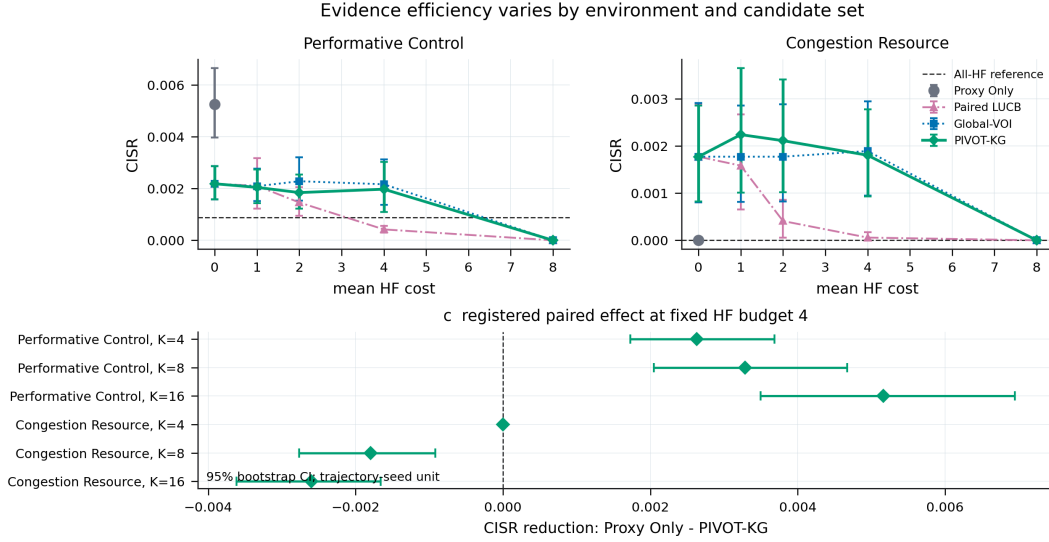


Figure 4: Evidence-efficiency frontier. Matched-cost curves and the fixed-budget forest report selection-regret reduction; the all-HF line is a reference. Intervals are seed bootstraps and no uniform ordering is asserted.

Table 2: Original HighwayEnv study and fresh-seed DEV replication, each with 60 test seeds.  $B$  counts candidate paired HF queries per decision. Positive contrasts favor PIVOT-KG; intervals are paired 95% bootstrap intervals.  $B = 2$  is primary and  $B = 4$  secondary in the replication.

| Cohort      | HF budget   | $n$ | Uniform ISR | PIVOT ISR | Contrast | 95% CI               |
|-------------|-------------|-----|-------------|-----------|----------|----------------------|
| Original    | 1           | 60  | 0.0492      | 0.0718    | -0.0226  | $[-0.0439, -0.0008]$ |
| Original    | 2 (primary) | 60  | 0.0422      | 0.0374    | 0.0049   | $[-0.0088, 0.0199]$  |
| Original    | 4           | 60  | 0.0144      | 0.0029    | 0.0115   | $[0.0008, 0.0237]$   |
| Replication | 2 (primary) | 60  | 0.0254      | 0.0261    | -0.0006  | $[-0.0143, 0.0145]$  |
| Replication | 4           | 60  | 0.0037      | 0.0004    | 0.0034   | $[-0.0007, 0.0099]$  |

calibration and post-decision truth audits are accounted for separately. Table 2 reports  $\text{ISR}_{\text{Uniform}} - \text{ISR}_{\text{PIVOT-KG}}$ , so positive values favor PIVOT-KG. This compares complete selection rules, including query allocation and calibrated estimates used for selection.

The original two-query contrast is unresolved ( $0.0049 [-0.0088, 0.0199]$ ); four queries are positive ( $0.0115 [0.0008, 0.0237]$ ), while one query is negative ( $-0.0226 [-0.0439, -0.0008]$ ). We then froze one replication with new, disjoint calibration/test cohorts,  $B = 2$  primary and  $B = 4$  secondary, without retuning or seed exclusions. The replication contrasts are  $-0.0006 [-0.0143, 0.0145]$  at  $B = 2$  and  $0.0034 [-0.0007, 0.0099]$  at  $B = 4$ ; both intervals cross zero. The replication does not establish a four-query advantage. This is external DEV evidence; the technical smoke exposure and reproduction protocol are disclosed in appendix A. The secondary budget cannot replace the primary conclusion or establish a general allocation guarantee.

## 6.7 STRESS TESTS BEYOND CONTROLLED ENVIRONMENTS

The public finance audit uses historical paths, virtual fills, and a participation/depth proxy. It has 0/7 depth-proxy reversals in the expanded primary sessions and 0/5 in the holdout; the pooled depth-minus-replay effect is approximately  $-4.16 \times 10^{-7}$  in native session units. This is a boundary diagnostic, not causal market impact. The strategic mechanisms are finite fixtures rather than general equilibrium, and external adaptive worlds are endpoint evidence rather than connected trajectories.

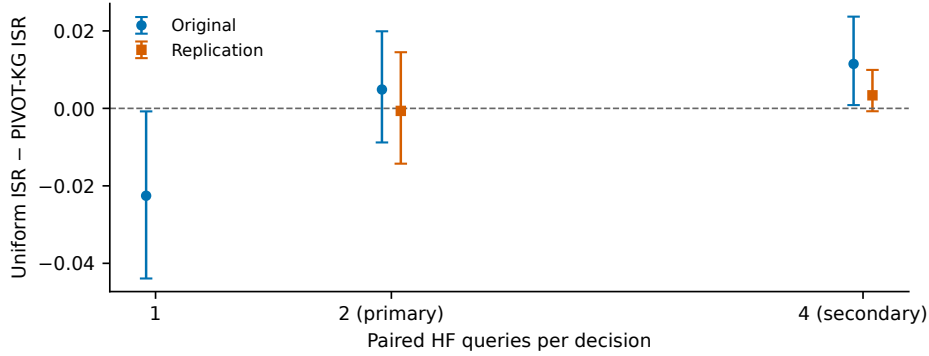


Figure 5: HighwayEnv paired allocation contrasts in two disjoint cohorts. Positive Uniform-minus-PIVOT ISR favors PIVOT-KG. Error bars are individual 95% bootstrap intervals over 60 seeds per cohort;  $B = 2$  is primary.

## 7 DISCUSSION AND SCOPE

The evidence supports a conditional claim. Deployment response can increase proxy–deployment error; an error changes the update only when it is large relative to the candidate margin; and PIVOT-KG can help when limited HF queries resolve that decision-relevant uncertainty. The sealed-query audit makes the last condition testable at the level of observable query criticality and residual separation, while stopping short of an unrecorded posterior or EVSI claim. A large response gap, a top-1 flip, or a positive result in one cohort is not by itself an allocation guarantee. The study does not claim universal superiority, simulator ground truth, market causality, or equilibrium behavior.

HighwayEnv reinforces the scope limit: the original one-query contrast is unfavorable, and the primary two-query comparison is unresolved in both cohorts. The original four-query advantage is also unresolved in the fresh-seed replication. The budget-dependent pattern is thus exploratory; these data establish neither a reproducible four-query advantage nor a budget-sufficiency mechanism.

## 8 CONCLUSION

An agent can be correct about the wrong world. The relevant statistical object for self-improvement is therefore the transition  $\pi \rightarrow \pi'$ , and the core question is whether its sign and ordering survive deployment-induced response. Across controlled and sealed worlds, deployment response separates proxy and deployment improvements, but the size of that separation does not by itself predict a ranking flip or an allocation gain. In the sealed study, 51 of 90 roots changed their proxy/deployment top-1, yet flipped roots did not have a larger normalized PIVOT–Uniform gain.

PIVOT-KG targets the next link in this chain: whether limited HF evidence can resolve a decision-relevant uncertainty before an update is promoted. Its positive Leduc contrast, near-null Kuhn result, and mixed or unresolved Melting Pot conditions support a conditional claim rather than universal superiority. The no-update action and paired evidence are part of the decision problem, so the method can retain the incumbent when all estimated improvements are weak or negative. The claims remain bounded by fixed response worlds, held-out sealed roots, and finite HF budgets; strategic fixtures are not equilibrium results, and the finance audit is not causal market evidence.

The broader implication is to validate consequences under the response induced by the proposed update. The resulting evidence ladder is

$$\text{observer} \longrightarrow \text{actor} \longrightarrow \text{strategic}, \quad (19)$$

Each arrow is an additional intervention, not an assumed truth claim; an HF budget matters only when it can change the update decision.

**AI Use Statement.** Generative AI tools were used for language editing, code and figure-formatting assistance, limited assistance in developing experimental infrastructure, and literature discovery. All experimental designs, mathematical claims, generated code, numerical results, cited sources, and manuscript claims were reviewed and verified by the authors, who take full responsibility for the work.

**Reproducibility Statement.** The anonymous supplement contains deterministic configurations, hash-indexed source evidence, figure tables, and rebuild commands. The public finance audit uses virtual fills and contains no credentials, private data, raw vendor archives, or live orders.

**Ethics Statement.** The study uses controlled virtual environments and public market data only; it involves no human subjects, personal data, live trading, or external authorization.

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## A EXTENDED FIGURES AND ARTIFACT NOTES

The release includes frozen source snapshots, result manifests, figure assets, source tables, and analysis scripts. The metadata records the unit of inference and distinguishes descriptive spans from bootstrap intervals.

**HighwayEnv replication protocol.** The original and replication cohorts each contain 20 calibration and 60 test roots, with no overlap between cohorts. The replication freezes the five-action panel, four lanes, 20 traffic vehicles, horizon 20, and covariance shrinkage 0.5. It refits the correction prior on its own calibration cohort before opening the full test cohort. The primary contrast is Uniform minus PIVOT-KG ISR at  $B = 2$ ;  $B = 4$  is secondary. We retain every registered root and use 4,000 paired bootstrap resamples with fixed RNG seeds. Individual 95% intervals condition on each fitted calibration model; they do not include calibration sampling uncertainty or form a simultaneous confidence statement. A technical smoke run exposed the first two replication test roots before the full run; those roots are retained without subsequent tuning or exclusions. This is therefore an external DEV replication, not a wholly blinded confirmatory test. The supplement includes the execution note, complete seed records, query and truth ledgers, dependency lock, frozen simulator code, and a standalone table/figure recomputation command. Calibration and post-decision truth audits are outside the logical query budget and are included in physical evaluation accounting. Uniform HF retains proxy estimates for unqueried candidates, whereas PIVOT-KG uses its calibrated posterior. The contrast therefore includes calibration and final selection as well as acquisition; it does not isolate an acquisition mechanism.

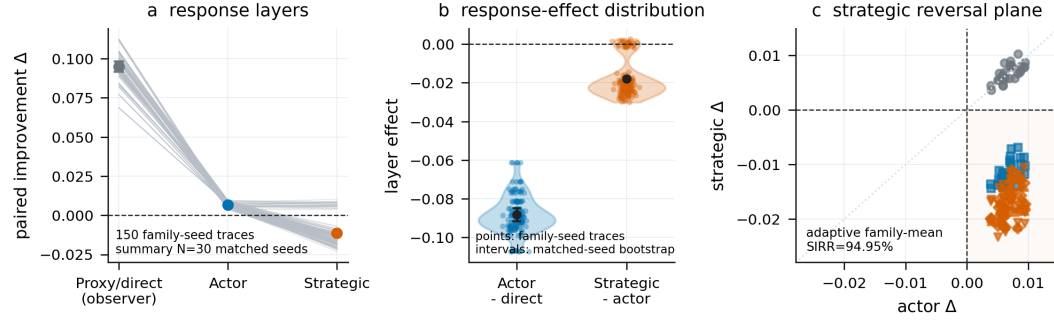


Figure 6: World-response decomposition. Panel a traces 150 family-by-seed observations (five opponent families  $\times$  30 matched seeds) through proxy/direct observer, actor, and strategic worlds; its summary intervals use the 30 matched seeds. Panel b separates mechanical and strategic effects; Panel c marks the strategic reversal quadrant.

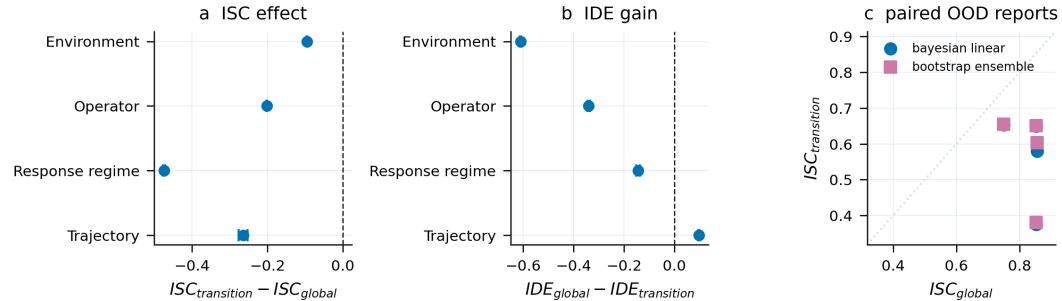


Figure 7: Powered OOD negative result. Forest horizontal ranges are descriptive spans across the two fitted model families in each split, not confidence intervals; the scatter retains the paired reports and the diagonal indicates equal ISC.

**Metric and claim boundary.** The canonical metric implementation is the frozen statistics module. The closed-loop CTI/CISR values sum over rounds; the budget frontier is a one-set ( $T = 1$ ) selection

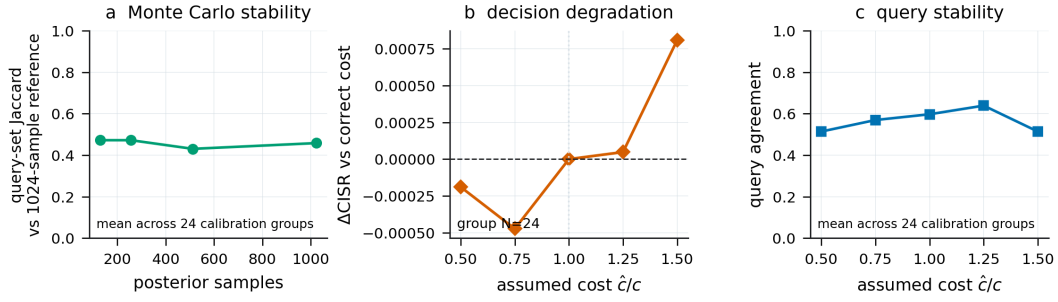


Figure 8: PIVOT-KG robustness to posterior sample count and cost misspecification. The curves report means over the registered calibration groups; they are numerical-stability diagnostics, not additional superiority claims.

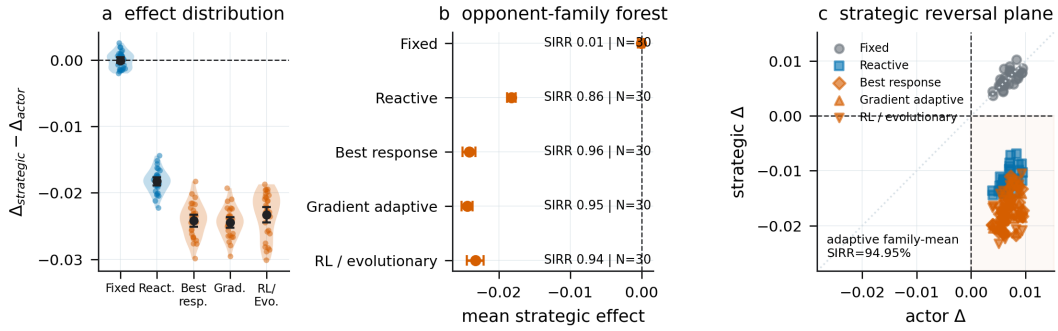


Figure 9: Strategic generalization across opponent mechanisms. Raincloud and forest views show family-specific effects and SIRR, each over 30 matched opponent-seed clusters. The reversal plane labels the unweighted mean over the three adaptive family-level SIRRs; it uses both color and marker shape so the failure is legible in grayscale.

problem. Their scales are therefore reported separately. The external adaptive and finance artifacts remain bounded evidence; no claim is promoted merely because a plot is visually strong. Internal experiment identifiers, configuration digests, and exact source hashes are retained in the supplement and figure metadata for audit. The raw sampled reversal-rate cells used in the operator-shift display remain available alongside the complete compressed transition stream.

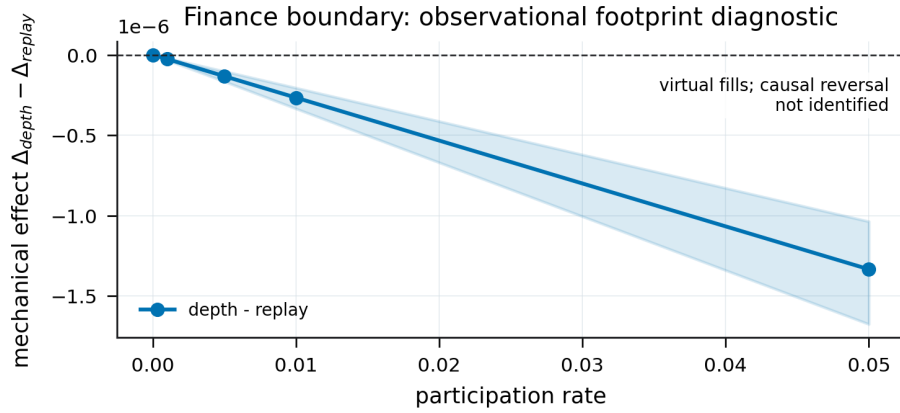


Figure 10: Initial single-asset public finance boundary diagnostic over participation. Shading is a seven-session descriptive interval; fills are virtual and the figure does not identify causal impact. The separate 12-session, three-asset expansion is summarized in the public-data table.

Table 3: Controlled and public evidence summary.

| Evidence layer         | Estimate | 95% interval       | Unit / interpretation                   |
|------------------------|----------|--------------------|-----------------------------------------|
| Operator shift         | 0.2627   | [0.1684, 0.3763]   | IDE contrast; paired transition         |
| Closed-loop validation | 1.4883   | [1.0618, 1.9105]   | CISR reduction; trajectory seed         |
| OOD evaluator contrast | -0.2590  | [-0.3588, -0.1707] | transition ISC minus global ISC; null   |
| Strategic adaptation   | -0.0240  | [-0.0249, -0.0231] | strategic minus actor; opponent cluster |

Table 4: Public-data boundary summary.

| Public-data boundary                  | Value     |
|---------------------------------------|-----------|
| Initial single-asset sessions         | 7         |
| Expanded primary sessions             | 12        |
| Expanded primary reversals / eligible | 0/7       |
| Expanded holdout reversals / eligible | 0/5       |
| Expanded pooled depth effect          | -4.16e-07 |
| Causal impact identified              | No        |